# Legitimacy Gating in Social Learning

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June 2026

**Abstract**

Standard models of social learning assume that reward feedback is always valid evidence: a bad outcome is information that one's belief was wrong. But people often respond to a bad outcome by questioning the rule that produced it rather than their own choice. We develop a computational social simulation of *legitimacy-gated* learning. Agents choose between rewarded actions, but rewards discipline beliefs only through a latent legitimacy state $L_{i,t}$ that measures whether the reward rule is accepted as evidence: the effective prediction error is $L_{i,t}$ times the raw prediction error. Legitimacy depends on perceived fairness, defined relative to a benchmark for a representative modal agent; because agents may misrepresent that benchmark, the same rule can be read as fair by one community and unfair by another. Fairness discomfort is asymmetric — disadvantageous unfairness hurts more than advantageous unfairness — and only discomfort beyond an acceptability threshold erodes legitimacy. Network geometry transmits both credibility and fairness benchmarks. Simulations identify four regimes: legitimate consensus, correct-but-illegitimate consensus, defensive wrong consensus, and fairness polarisation. The central finding is that legitimacy collapse converts a *correctable* early error into a persistent one: a society that would have corrected an early mistake (93% optimal-arm mass) instead locks onto it (1%) once corrective feedback loses authority. The model builds on the credibility-transmission mechanism of Bontemps and Banerjee (2026) but studies a distinct behavioural layer — whether feedback is accepted as legitimate evidence — and is self-contained. A replication package regenerates every figure and table.

**Keywords:** social learning; legitimacy; fairness; reinforcement learning; reward discounting; community networks; agent-based simulation; polarisation.

## 1 Introduction

A bad outcome is ambiguous. It can be read as evidence that one's belief was wrong, or as evidence that the rule assigning outcomes is unfair. Models of social and reinforcement learning almost always assume the first reading: realised rewards are valid evidence and prediction errors discipline beliefs directly (Sutton and Barto, 2018; DeGroot, 1974; Golub and Jackson, 2010). Yet a large body of work on fairness, legitimacy and motivated cognition documents the second: people discount, reinterpret or reject information that conflicts with what they feel they deserve (Rabin, 1993; Fehr and Schmidt, 1999; Tyler, 2006; Kunda, 1990; Festinger, 1957). This paper asks a simple question: *when do agents stop treating reward feedback as valid evidence because the reward rule is perceived as unfair, and how does network geometry shape the result?*

We answer it with a computational social simulation in which the authority of feedback is itself a learned, social quantity. Each agent carries a latent *legitimacy* state $L_{i,t} \in [0, 1]$ that gates learning: the effective prediction error is $L_{i,t}$ times the raw prediction error, so that when the rule is seen as illegitimate, even strong corrective feedback barely moves beliefs. Legitimacy is not exogenous. It falls when accumulated *fairness discomfort* exceeds an acceptability threshold

and recovers slowly when outcomes are acceptable. Fairness is judged against a benchmark for a representative *modal agent*; agents may hold distorted benchmarks, and those benchmarks are themselves transmitted on the network. Discomfort is asymmetric: disadvantageous unfairness weighs more than advantageous unfairness.

The point of departure is a credibility-transmission baseline in which confidence is generated by the decision process and socially transmitted as credibility (Bontemps and Banerjee, 2026). There, agents can become collectively wrong, but they never dispute the rule: rewards remain valid evidence. The present paper relaxes exactly that assumption. The behavioural layer it adds is not who is believed (credibility) but whether the feedback counts at all (legitimacy). The two interact: credibility-weighted transmission still spreads actions and confidence, while a second channel spreads fairness benchmarks, and legitimacy decides how much realised reward is allowed to teach.

The simulations identify four regimes. In *legitimate consensus* agents converge on the reward-maximising action with high legitimacy. In *correct-but-illegitimate consensus* they learn the optimal action but legitimacy collapses: they know what works yet believe the game is unfair. In *defensive wrong consensus* they lock onto the wrong action and discount the corrective feedback that would overturn it. In *fairness polarisation* communities differ in legitimacy — and even in action — under an identical reward rule, purely because they hold different fairness benchmarks. The headline result is that legitimacy gating makes wrong consensus *persistent*: a configuration that corrects an early error when feedback is fully authoritative instead entrenches it once legitimacy collapses.

**Contributions.**

(i) A self-contained agent-based model in which reward feedback teaches only through an endogenous legitimacy gate, with fairness judged against a transmissible modal-agent benchmark and asymmetric discomfort.

(ii) Four operational regimes — legitimate, correct-but-illegitimate, defensive-wrong, and fairness-polarised — and an analytical legitimacy-collapse condition.

(iii) Six experiments showing asymmetric legitimacy loss, an acceptability threshold, benchmark-driven fairness polarisation without payoff differences, legitimacy-gated persistence of wrong consensus, and the role of network geometry.

(iv) A replication package (code, tests, configs, scripts) regenerating every result.

## 2 Background and positioning

The model sits between three literatures. *Social and reinforcement learning* explains how agents aggregate dispersed information and learn from reward (DeGroot, 1974; Golub and Jackson, 2010; Sutton and Barto, 2018), and opinion-dynamics models explain when interaction yields consensus, clustering or polarisation (Hegselmann and Krause, 2002; Flache et al., 2017). These models treat the evidential status of feedback as fixed. *Fairness and inequity aversion* in economics formalises the idea that agents care about how outcomes compare to a reference, asymmetrically (Rabin, 1993; Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000; Charness and Rabin, 2002); we borrow the asymmetric reference-dependent form but place it inside a learning loop rather than a one-shot utility. Crucially, fairness ideals are *heterogeneous* — individuals hold systematically different notions of a fair reference (Cappelen et al., 2007; Andre, 2024) — which is the empirical basis for our transmissible, community-specific benchmark $\hat{M}$ and the fairness-polarisation regime it generates. Legitimacy is thus an *endogenous, socially learned norm*,

in the tradition of endogenous preferences and culturally transmitted norms (Bowles, 1998; Tyler, 2006). *Legitimacy and motivated cognition* explain why perceived unfairness changes whether a rule is accepted and whether discordant information is processed (Tyler, 2006; Kunda, 1990; Festinger, 1957); we operationalise this as a gate on the prediction error. Neuroeconomically, the gate is a modulation of the evidential weight of reward prediction errors: social and non-social learning share prediction-error machinery, and self- versus other-referenced prediction errors are represented and can be up- or down-weighted according to context (Joiner et al., 2017). The legitimacy state is exactly such a context variable, scaling how much a realised reward is allowed to teach. This places the mechanism inside the current debate over what orbitofrontal and prefrontal cortex compute — explicit economic value (Padoa-Schioppa and Conen, 2017) versus a latent task-state or cognitive map that contextualises value updating (Hayden and Niv, 2021); legitimacy is a state-like context variable of the second kind. Behaviourally, a *defensive-wrong* consensus is a failure of *reversal learning* — the capacity to overturn a previously reinforced contingency, whose neural substrate is well characterised (Izquierdo et al., 2017) — brought about socially rather than by lesion. We connect the resulting regimes to behavioural-equilibrium theory (Section 6): quantal response and its regular, axiomatic form (McKelvey and Palfrey, 1995; Goeree et al., 2005), the systematic experimental anomalies these were built to explain (Goeree and Holt, 2001), the set-valued M-equilibrium (Goeree and Louis, 2021), and nonequilibrium / level-$k$ reasoning about a modal agent (Nagel, 1995; Camerer et al., 2004; Crawford et al., 2013); the learning dynamics themselves are of the reinforcement / experience-weighted family studied in learning-in-games (Erev and Roth, 1998; Camerer and Ho, 1999).

Relative to the credibility-transmission baseline (Bontemps and Banerjee, 2026), the contribution is a different behavioural layer. There, the open question is whose action is believed; here it is whether the feedback is admitted as evidence at all. We re-use the credibility backbone (process-based choice, endogenous confidence, dual-channel social influence on community networks) but define every component independently, and the new mechanism — legitimacy gating with a transmissible fairness benchmark — is original to this paper.

## 3 The model

Figure 1 summarises the mechanism. We describe it in ODD+D terms in Appendix B.

**Environment and states.** Time is discrete, $t \in \mathbb{N}_0$. Agents $V = \{1, \dots, N\}$ are partitioned into $M$ communities with row-stochastic coupling matrix $B$ (cohesion $B_{cc}$, permeability $\Gamma_c = 1 - B_{cc}$); the induced agent network has balanced block weights $W_{ij} = B_{cd}/N_d$. Two arms pay Bernoulli rewards with means $\mu_1, \mu_2$; $a^\star = \arg\max_a \mu_a$. Each agent carries values $Q_{i,t}(a)$, confidence $C_{i,t}$, legitimacy $L_{i,t} \in [0,1]$, and a perceived modal-agent fairness benchmark $\hat{M}_{i,t} \in [0,1]$.

**Choice, confidence, credibility.** The anticipatory credibility field is the confidence-weighted mass of neighbours' previous actions, $S_{i,t}(a) = \sum_d B_{cd}\, \phi_{d,t-1}(a)$ with $\phi_{c,t}(a) = \frac{1}{N_c} \sum_{i \in V_c} \mathbf{1}\{A_{i,t} = a\} C_{i,t}$. Augmented values are $V_{i,t}(a) = Q_{i,t}(a) + \lambda S_{i,t}(a)$, the drift is $v_{i,t} = \beta(V_{i,t}(1) - V_{i,t}(2))$, and the action follows the exact symmetric-boundary hitting probability $\mathbb{P}(A_{i,t} = 1) = (1 + e^{-2v_{i,t}a/\sigma^2})^{-1}$. Confidence is the balance of evidence, $C_{i,t} = |2\mathbb{P}(A_{i,t} = 1) - 1| \in (0,1)$.

**Modal-agent fairness benchmark.** Fairness is judged against a representative *modal agent*. We distinguish three objects: the true modal-agent benchmark $M$; agent $i$'s *perceived* benchmark $\hat{M}_{i,t} \in [0,1]$; and the perceived fair reward for an action, $r^F_{i,t}(a) = f(a, \hat{M}_{i,t})$. The baseline uses the

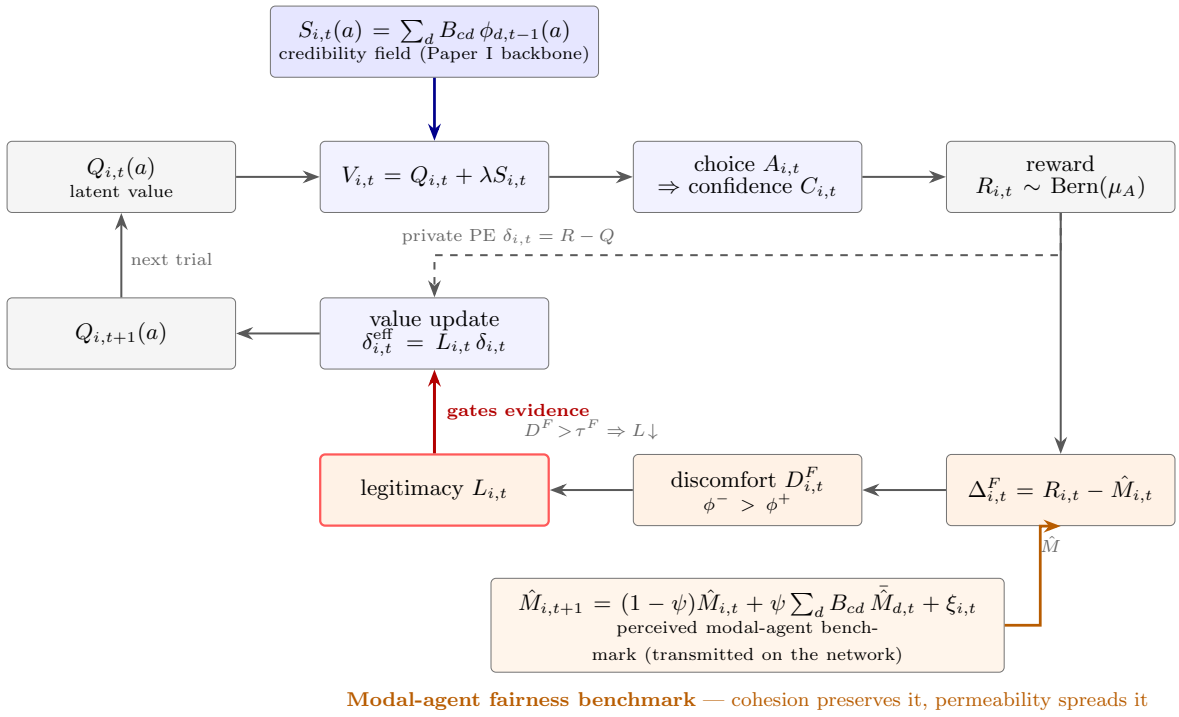



Figure 1: Mechanism. The value/choice pipeline (top) is the credibility backbone: augmented values drive a process-based choice and endogenous confidence, and a credibility field transmits confidence-weighted actions. The reward then enters a fairness branch (right): the discrepancy from a perceived modal-agent benchmark $\hat{M}$ produces asymmetric discomfort, which moves legitimacy $L$ across an acceptability threshold $\tau^F$. Legitimacy *gates* the prediction error ($\delta^{\mathrm{eff}} = L\,\delta$, red), so illegitimate feedback barely teaches. The modal-agent benchmark is itself transmitted on the network (bottom).

scalar form $f(a, \hat{M}) = \hat{M}$; an effort–reward form $f(a, \hat{M}) = b_0 + b_1 \mathbf{1}\{a = \text{high effort}\} + b_2 \hat{M}$, in which a high-effort action is expected to deserve more reward, is available behind a configuration flag and leaves the baseline experiments unchanged. $\hat{M}_{i,t}$ is the perceived benchmark reward entitlement of the modal agent under the current rule; agents may *misrepresent* it ($\hat{M}_{i,t} \neq M$), and the perceived benchmark is itself socially learned (Eq. 6).

**Fairness discrepancy, discomfort, legitimacy.** The fairness discrepancy is

$$\Delta^F_{i,t} = R_{i,t} - r^F_{i,t}(A_{i,t}), \tag{1}$$

negative for disadvantageous and positive for advantageous unfairness. Discomfort is asymmetric,

$$D^F_{i,t} = \phi^-[-\Delta^F_{i,t}]_+ + \phi^+[\Delta^F_{i,t}]_+, \qquad \phi^- > \phi^+ \geq 0, \tag{2}$$

and only discomfort beyond an acceptability threshold $\tau^F$ erodes legitimacy, which otherwise recovers slowly:

$$L_{i,t+1} = \Pi_{[0,1]}\Big(L_{i,t} - \nu_F\,[D^F_{i,t} - \tau^F]_+ + \nu_R\,\mathbf{1}\{D^F_{i,t} \leq \tau^F\}\Big). \tag{3}$$

**Legitimacy-gated learning.** The raw prediction error is $\delta_{i,t} = R_{i,t} - Q_{i,t}(A_{i,t})$ and the *effective* prediction error is gated by legitimacy,

$$\delta^{\text{eff}}_{i,t} = L_{i,t}\,\delta_{i,t}. \tag{4}$$

Values are revised by the gated private error plus a retrospective social error (neighbours' observed outcomes, weighted by their confidence through $g(\bar{C}) = \gamma\bar{C}^{\omega}$):

$$Q_{i,t+1}(a) = \Pi_{[0,1]}\Big(Q_{i,t}(a) + \mathbf{1}\{A_{i,t} = a\}\,\alpha\,\delta^{\text{eff}}_{i,t} + \eta\,g(\bar{C}_{i,t}(a))\,\delta^{\text{soc}}_{i,t}(a)\Big). \tag{5}$$

All prediction errors and the gate use the pre-update state; updates are applied simultaneously.

**Modal-agent transmission.** Fairness benchmarks diffuse on the same community geometry,

$$\hat{M}_{i,t+1} = \Pi_{[0,1]}\Big((1-\psi)\hat{M}_{i,t} + \psi\sum_{d} B_{cd}\,\bar{\hat{M}}_{d,t} + \xi_{i,t}\Big), \tag{6}$$

so high cohesion preserves local benchmarks and high permeability spreads them.

**When legitimacy collapses.** Equation (3) gives a transparent collapse condition.

**Proposition 1** (Legitimacy drift)**.** *Under a stationary choice/outcome distribution, the expected per-trial change in legitimacy (away from the bounds) is* $\mathbb{E}[\Delta L] = -\nu_F\,\mathbb{E}[(D^F - \tau^F)_+] + \nu_R\,\mathbb{P}(D^F \le \tau^F)$*. Legitimacy drifts downward (collapses) iff* $\nu_F\,\mathbb{E}[(D^F - \tau^F)_+] > \nu_R\,\mathbb{P}(D^F \le \tau^F)$.

This makes the roles of the acceptability threshold $\tau^F$, the benchmark $\hat{M}$ (through $D^F$), and the loss/recovery rates explicit, and anticipates the threshold sweep of Section 5.

Two further properties make the two signatures of the mechanism precise.

**Proposition 2** (Evidence discounting slows correction)**.** *Along a trajectory on which legitimacy varies slowly around a level* $\bar{L}$*, the gated private update (4)–(5) is a reinforcement learner with effective step size* $\alpha\bar{L}$*: the chosen-arm value relaxes toward its reward mean with per-visit modulus* $1 - \alpha\bar{L}$*, so the time to overturn an incorrect value gap scales like* $1/(\alpha\bar{L})$ *and diverges as* $\bar{L} \to 0$*. Legitimacy collapse therefore makes a wrong lead persistent rather than correctable.*

**Lemma 1** (Benchmark-gap decay)**.** *In the deterministic two-community mean field with balanced block weights, the between-community benchmark gap* $\Delta\bar{\hat{M}}_t = \bar{\hat{M}}_{1,t} - \bar{\hat{M}}_{2,t}$ *obeys* $\Delta\bar{\hat{M}}_{t+1} = (1 - 2\psi\Gamma)\,\Delta\bar{\hat{M}}_t$*, decaying geometrically at rate* $2\psi\Gamma$*; a persistent gap (fairness polarisation) requires* $\psi\Gamma$ *small.*

Proofs of Propositions 1, 2, Lemma 1 and Proposition 3 are collected in Appendix A. Proposition 2 is the mechanism behind defensive wrong consensus (Section 5.2) and Lemma 1 behind the fairness-polarisation region of the secondary phase diagram (Section 5.1); the four qualitative outcomes are organised in Figure 2 by action correctness and legitimacy.

## 4 Simulation design

**Information structure.** Before choosing, agents observe neighbours' previous actions and confidence signals (the credibility field). After outcomes are realised, they observe neighbours' actions, realised rewards and confidence for the retrospective channel, and they observe their own realised reward relative to the perceived modal-agent benchmark. Confidence is communicated or

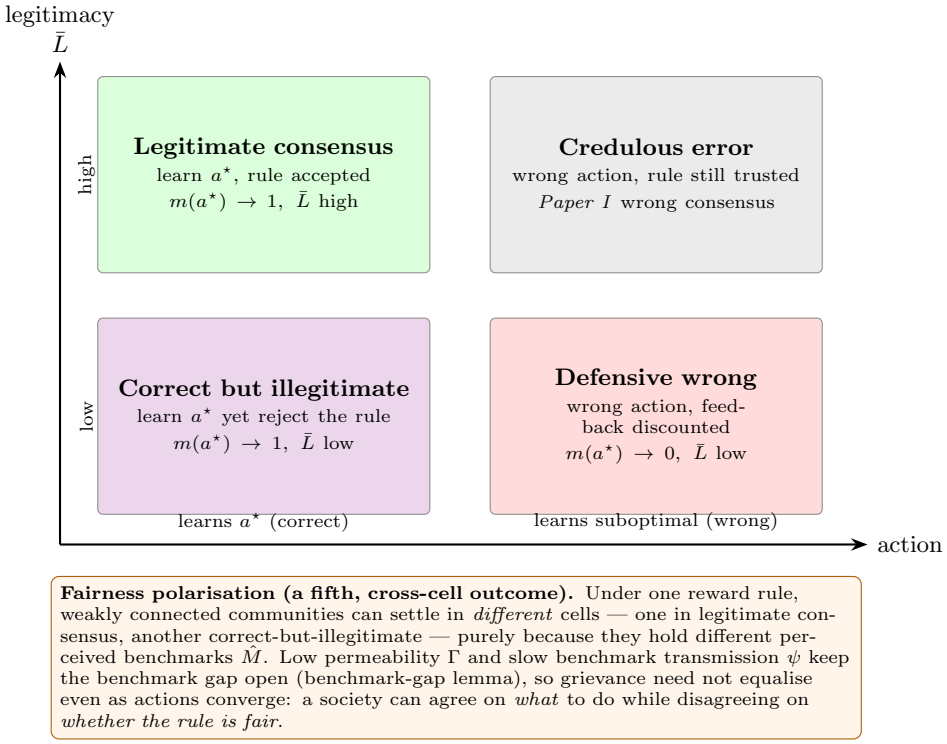



Figure 2: Regime map. The two collective achievements — learning the optimal action and accepting the rule as legitimate — are independent axes. Legitimate consensus and defensive wrong consensus lie on one diagonal; correct-but- illegitimate consensus and credulous error (the wrong consensus of the credibility baseline) on the other. Fairness polarisation is communities occupying different cells under the same rule.

behaviourally inferred subjective reliability, not objective accuracy; the modal-agent benchmark is a perceived entitlement, not the true rule.

We use two communities, binary actions, stationary rewards, balanced block weights, and a synchronous simultaneous update. Observables include community action mass $m_{c,t}(a)$, mean legitimacy $\bar{L}_{c,t}$, the perceived benchmark $\hat{\bar{M}}_{c,t}$, the acceptability rate, the evidence-discount gap $|\delta| - |\delta^{\text{eff}}|$, the legitimacy gap $G_t^L = |\bar{L}_{1,t} - \bar{L}_{2,t}|$, and the modal-agent gap $G_t^M$. Terminal outcomes are classified as *legitimate consensus* (both communities place $\geq \theta$ on $a^\star$ and mean legitimacy is high), *correct-but-illegitimate* (optimal-arm consensus, low legitimacy), *defensive wrong* (suboptimal-arm consensus, low legitimacy), *fairness polarisation* (communities differ in legitimacy or action), or unresolved. Regime probabilities are Monte Carlo frequencies; the implementation passes the verification suite of Appendix C before any experiment is run. Default and per-experiment parameters are in Appendix D; the replication package is authoritative. Table 1 positions the model and Table 2 lists the variables.

**Uncertainty reporting.** Regime probabilities are empirical frequencies over independent replications, reported with Wilson 95% intervals; means carry Monte Carlo standard errors and percentile-bootstrap 95% intervals. The ablations (Section 5.2) use common random numbers — the same seed, hence the same exogenous reward and decision-noise environment, across variants, with realised rewards following each variant's action path — so the full-vs-ablation paired difference identifies a component's effect, reported with a paired-bootstrap 95% interval.

Table 1: Positioning relative to adjacent model families.

| Tradition | Typical mechanism | Feedback authority | What this paper changes |
|---|---|---|---|
| Social/RL learning | reward-driven value update | fixed (always valid) | legitimacy gate on evidence |
| Inequity aversion | reference-dependent utility | one-shot, no learning | fairness inside a learning loop |
| Legitimacy / proc. justice | rule acceptance | exogenous | endogenous, socially learned |
| Opinion dynamics | weighted averaging | no reward rule | reward legitimacy as state |
| Credibility (Paper I) | confidence-weighted transmission | valid evidence | feedback can be delegitimised |

Table 2: Model variables and interpretations.

| Symbol | Interpretation |
|---|---|
| $Q_{i,t}(a)$ | latent action value |
| $C_{i,t}$ | endogenous confidence (balance of evidence) |
| $S_{i,t}(a)$ | credibility-weighted social signal |
| $L_{i,t}$ | legitimacy of the reward rule (gates evidence) |
| $\hat{M}_{i,t}$ | perceived modal-agent fairness benchmark |
| $M$ | true modal-agent benchmark |
| $\Delta^F_{i,t}$ | fairness discrepancy $R - r^F$ |
| $D^F_{i,t}$ | asymmetric fairness discomfort |
| $\delta^{\text{eff}}_{i,t}$ | legitimacy-gated prediction error $L\,\delta$ |

# 5 Results

**II.1 Agents mind unfairness more when it harms them.** Holding the absolute fairness deviation fixed, disadvantageous unfairness ($\Delta^F < 0$) erodes legitimacy far more than advantageous unfairness ($\Delta^F > 0$), because $\phi^- > \phi^+$ pushes disadvantageous discomfort across the acceptability threshold while advantageous discomfort stays below it (Figure 3). This asymmetry is the engine of the regimes below: under-reward is delegitimising, over-reward is tolerated.

**II.2 An acceptability threshold separates tolerance from collapse.** Sweeping $\tau^F$ at a fixed level of unfairness, terminal mean legitimacy rises from near 0 to near 1 (Figure 4c): a low threshold makes unfairness delegitimising, a high threshold makes it tolerable, with a steep transition between. This is the empirical counterpart of Proposition 1.

**II.3 Fairness polarisation without payoff differences.** Two communities under the *same* reward rule, but with opposed modal-agent benchmarks ($\hat{M}_1 = 0.25$, $\hat{M}_2 = 0.75$) and low permeability, polarise in legitimacy: the low-entitlement community keeps $\bar{L} \approx 1$ while the high-entitlement community falls to $\bar{L} \approx 0.37$, a terminal gap $G^L \approx 0.63$ (Figure 4a,b). Because benchmarks transmit slowly under low permeability, the difference persists. Identical material payoffs thus produce divergent legitimacy purely through divergent fairness benchmarks.

**II.4 Legitimacy gating makes wrong consensus persistent.** This is the central result (Figure 5a). At a moderate transmission strength with an early lead on the suboptimal arm, the *ungated* model self-corrects: terminal optimal-arm mass $m(a^\star) = 0.93$ (legitimate consensus).

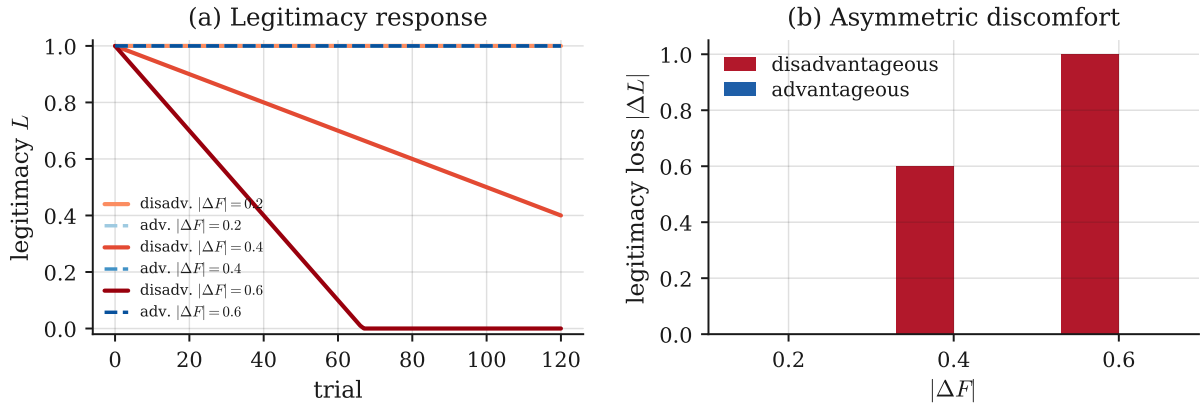



Figure 3: II.1. (a) Legitimacy trajectories under a fixed fairness deviation, by sign and magnitude. (b) Legitimacy loss is large for disadvantageous and negligible for advantageous unfairness.

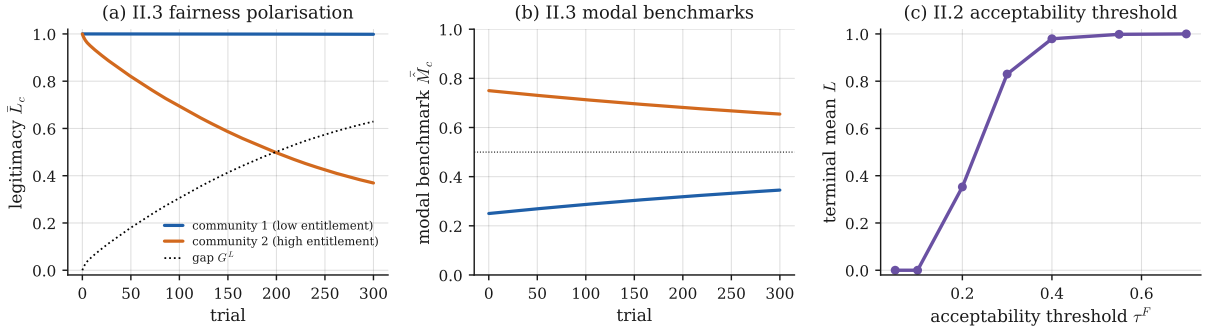



Figure 4: (a) II.3 community legitimacy and the gap $G^L$; (b) II.3 modal-agent benchmarks remain split under low permeability; (c) II.2 terminal mean legitimacy versus the acceptability threshold $\tau^F$.

Switching on the legitimacy gate, with the benchmark set so that the poorly paying arm feels unfair, legitimacy collapses while agents are still on the wrong arm; corrective feedback then loses authority ($\delta^{\text{eff}} = L\delta \to 0$), and the same configuration locks onto the wrong action: $m(a^\star) = 0.01$, $\bar{L} = 0.01$ — defensive wrong consensus. Adding modal-agent misrepresentation leaves the outcome wrong and lowers legitimacy further. Negative evidence loses its disciplining power exactly when it is most needed.

**II.5 Network geometry tunes legitimacy and its polarisation.** Varying cross-community permeability $\Gamma$ on the misrepresentation setup, low permeability sustains a large legitimacy gap $G^L$ (fairness polarisation) while high permeability mixes benchmarks toward a common value and a common legitimacy level (Figure 6). Permeability that would spread corrective information in a pure credibility model here also spreads fairness benchmarks, so its effect on collective legitimacy is not monotone in welfare.

**II.6 Epistemic success and legitimacy failure can diverge.** When the optimal arm is learnable but the entitlement benchmark is high, agents learn what works yet stop believing the rule is fair: $m(a^\star) \to 1$ while mean legitimacy decays to $\approx 0.01$ (Figure 5b). Correct beliefs and accepted rules are distinct collective achievements.

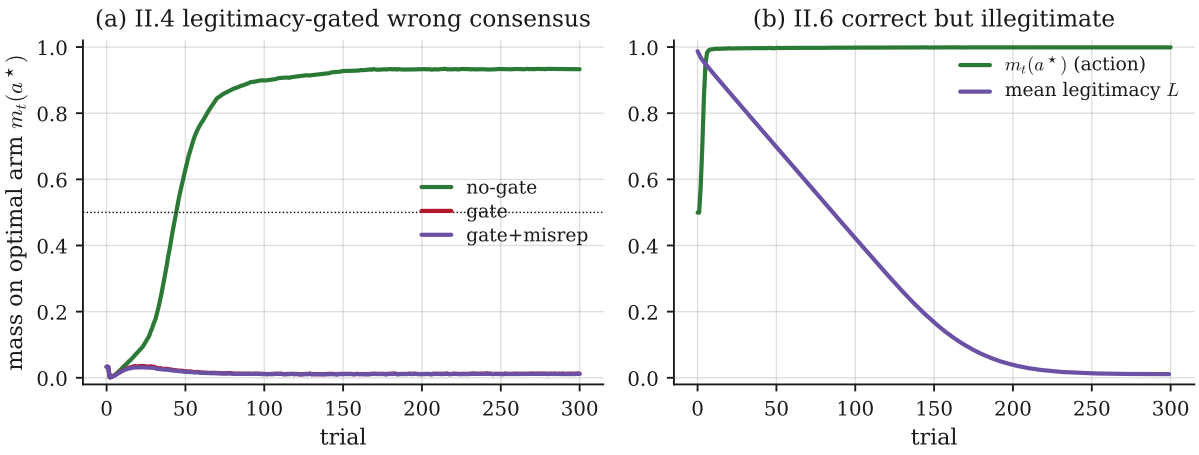



Figure 5: (a) II.4: with the legitimacy gate off the society corrects an early wrong lead ($m(a^\star) \to 0.93$); with the gate on it locks onto the wrong arm ($\to 0.01$). (b) II.6: agents learn the optimal action ($m(a^\star) \to 1$) while mean legitimacy decays to near zero — correct but illegitimate.

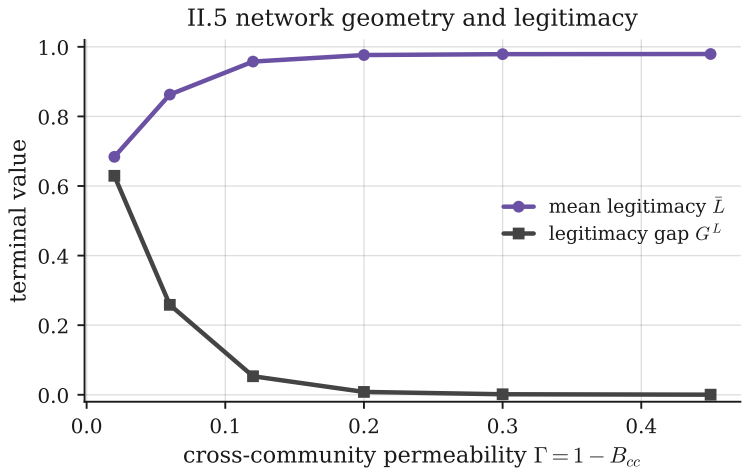



Figure 6: II.5. Terminal mean legitimacy $\bar{L}$ and the legitimacy gap $G^L$ versus permeability $\Gamma$.

### 5.1 Phase diagrams

Figure 7 maps the regimes. The *main* phase diagram sweeps the acceptability threshold $\tau^F$ against the initial benchmark disagreement $|\Delta\hat{M}_0|$ (with a learnable optimum, so action consensus forms and defensive-wrong does not arise in this slice). Three regimes tile the plane: a high-$\tau^F$ band of legitimate consensus; a low-$\tau^F$ band of correct-but-illegitimate consensus (both communities delegitimise); and a fairness-polarisation wedge at intermediate $\tau^F$ with large benchmark disagreement, where the high-entitlement community delegitimises while the low-entitlement one does not. The *secondary* diagram shows that the legitimacy gap $G^L$ is large only when permeability $\Gamma$ *and* modal transmission $\psi$ are both small: fairness polarisation requires that distorted benchmarks neither mix across communities nor average within them. Geometry that spreads information also dissolves grievance.

### 5.2 Which components are necessary? Ablations

At the legitimacy-gated wrong-consensus operating point ($R = 200$, common random numbers), the full model produces defensive wrong consensus with probability 1.00. Figure 8 and Table 4

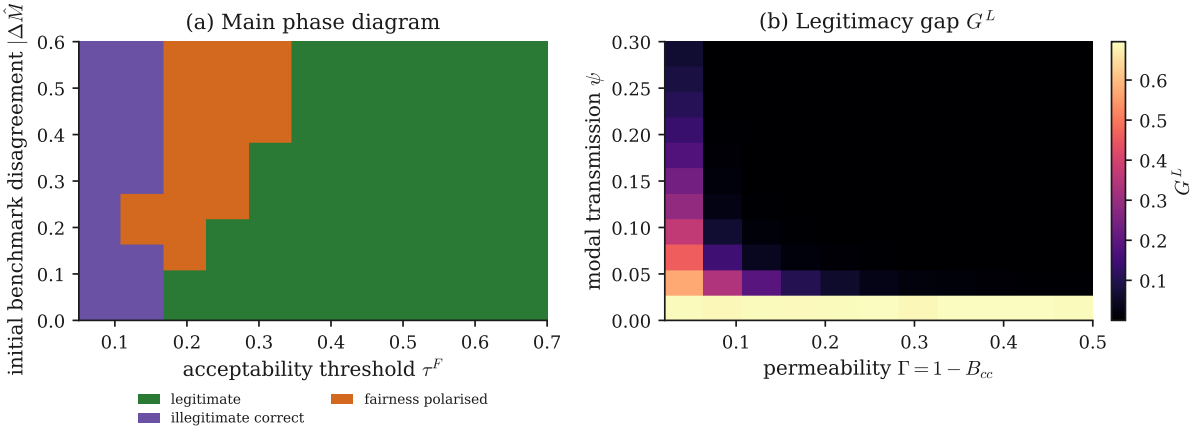



Figure 7: Phase diagrams ($R = 60$ per cell, $11 \times 11$). (a) terminal regime over $\tau^F$ (acceptability) and $|\Delta\hat{M}_0|$ (initial benchmark disagreement). (b) legitimacy gap $G^L$ over permeability $\Gamma$ and modal transmission $\psi$.

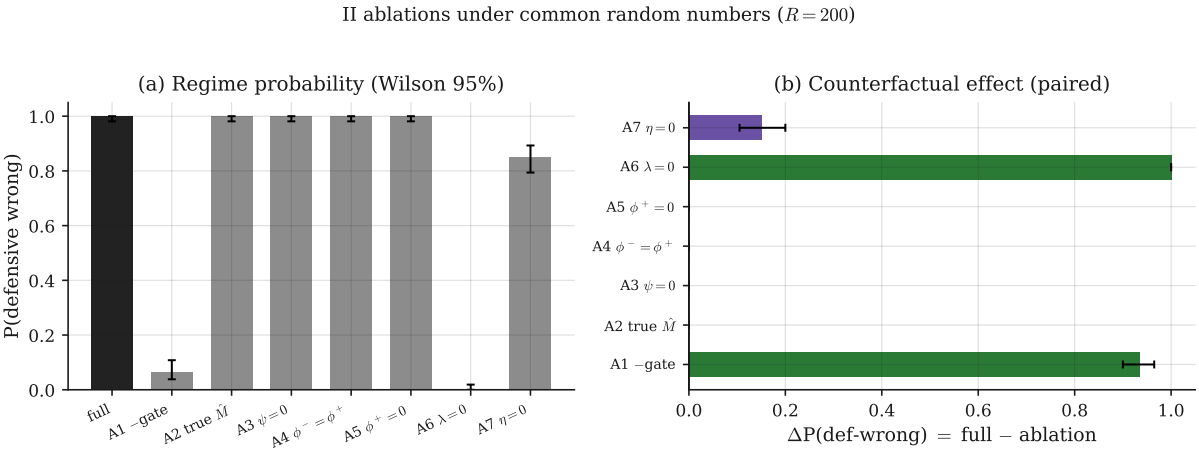



Figure 8: Ablations under common random numbers ($R = 200$). (a) P(defensive wrong) per variant with Wilson 95% intervals; (b) paired counterfactual effect $\Delta$ =P(def.-wrong|full)−P(def.-wrong|ablation) with paired-bootstrap 95% intervals. Green: necessary; purple: contributory; grey: not necessary.

report the paired counterfactual effects. Two components are *necessary*: removing the legitimacy gate (A1) restores correction ($m(a^\star)$ : $0.01 \to 0.91$, $\Delta = +0.93$), and removing the credibility field (A6) prevents the wrong *action* from locking in ($m(a^\star) \to 0.79$) although legitimacy still collapses ($\bar{L} \to 0.01$) — the gate alone yields correct-but-illegitimate, not defensive-wrong. The retrospective channel (A7) is *contributory* ($\Delta = +0.15$). Modal-agent misrepresentation and the discomfort asymmetry (A2–A5) are *not* necessary for defensive wrong consensus at this point — they instead drive the fairness-polarisation regime. Defensive wrong consensus is thus the conjunction of legitimacy collapse (the gate) and credibility-weighted amplification (the field).

## 6 Equilibrium interpretation

The model is a dynamic learning process, not an equilibrium solved each period, but its absorbing regimes have clean readings in behavioural game theory.

*A mean-field phase portrait.* The model admits a deterministic large-population reduction (`legitimacy/meanfield.py`): with a scalar benchmark $\hat{M}$ held fixed, the state collapses to the

Table 3: Experiment outcomes (standard scale). $m(a^\star)$ is terminal optimal-arm mass; $\bar{L}$ is terminal mean legitimacy; $G^L$ is the between-community legitimacy gap. Scenario parameters: `code/legitimacy/scenarios.py`.

| Experiment | Manipulation | Outcome |
|---|---|---|
| II.1 adv. vs disadv. | $\Delta^F = \pm x$, $\lvert x\rvert$ fixed | legitimacy loss 1.00 (disadv.) vs 0.00 (adv.) |
| II.2 acceptability | sweep $\tau^F$ | mean $L$ rises $0.05 \to 1.00$ across $\tau^F$ |
| II.3 modal misrep. | $\hat{M}_1 \neq \hat{M}_2$ | fairness polarisation, $G^L = 0.63$ |
| II.4 no gate | legitimacy off | $m(a^\star) = 0.93$, legitimate consensus |
| gate | legitimacy on | $m(a^\star) = 0.01$, $\bar{L} = 0.01$: **defensive wrong** |
| II.6 correct/illegit. | high entitlement | $m(a^\star) = 1.00$, $\bar{L} = 0.01$ |

Table 4: Ablation effects at the legitimacy-gated wrong-consensus operating point, $R = 200$ under common random numbers. P(def.-wrong) carries a Wilson 95% interval; $\Delta$ is the paired counterfactual effect full−ablation with a paired-bootstrap 95% interval.

| Variant | P(def.-wrong) [95%] | $m(a^\star)$ | $\Delta$P(def.-wrong) [95%] |
|---|---|---|---|
| full | 1.00 [0.98, 1.00] | 0.01 | — |
| A1 no gate | 0.07 [0.04, 0.11] | 0.91 | +0.94 [+0.90, +0.96] |
| A2 true $\hat{M}$ | 1.00 [0.98, 1.00] | 0.01 | +0.00 [+0.00, +0.00] |
| A3 $\psi$=0 | 1.00 [0.98, 1.00] | 0.01 | +0.00 [+0.00, +0.00] |
| A4 $\phi^-$=$\phi^+$ | 1.00 [0.98, 1.00] | 0.01 | +0.00 [+0.00, +0.00] |
| A5 $\phi^+$=0 | 1.00 [0.98, 1.00] | 0.01 | +0.00 [+0.00, +0.00] |
| A6 $\lambda$=0 | 0.00 [0.00, 0.02] | 0.79 | +1.00 [+1.00, +1.00] |
| A7 $\eta$=0 | 0.85 [0.79, 0.89] | 0.17 | +0.15 [+0.10, +0.20] |

value contrast $\Delta = Q(a^\star) - Q(\text{other})$ and the shared legitimacy $L$, with choices following the exact quantal (logistic) map and rewards, the gated update and the legitimacy update taken in expectation. Motivated cognition makes legitimacy a *self-reinforcing attitude* rather than a passively eroding quantity: a legitimate stance disposes agents to read outcomes charitably, an illegitimate one to read them as further grievance (Kunda, 1990; Festinger, 1957). We encode this as a tilted double well (cusp) for $x = L - \frac{1}{2}$,

$$\dot{x} = r\,x - s\,x^3 + h(\hat{M}), \qquad h(\hat{M}) = -\mu_t(\bar{D}^F(\hat{M}) - \tau^F), \tag{7}$$

where the tilt $h$ decreases in perceived unfairness $\bar{D}^F - \tau^F$ and $r > 0$ is the strength of the self-reinforcement. The constants are *calibrated from the model* (`calibrate_cusp`): the tilt vanishes at the discomfort-balance benchmark $\hat{M}^\star$ (where $\bar{D}^F = \tau^F$); the two wells are pinned to the model's birth-death legitimacy levels $\nu_R a/(\nu_R a + \nu_F e)$ — with acceptance rate $a$ and expected excess discomfort $e$ from the fairness rates $\nu_R, \nu_F, \phi^\pm, \tau^F$ — evaluated at the fair and unfair edges of the fold band; the tilt gain then places the two folds symmetrically about $\hat{M}^\star$. Only the self-reinforcement strength has no baseline counterpart. Its rest points are the (choice, legitimacy) equilibria; the flow is the phase portrait of Figure 9. Because $\Delta$ relaxes toward the correct contrast whenever $L > 0$, the value coordinate is monostable, but legitimacy is bistable: for a *fair* rule the plane has one interior attractor at high legitimacy (legitimate consensus), for an *unfair* rule one interior attractor at low legitimacy (correct-but-illegitimate), and for *intermediate* entitlement *two* interior attractors separated by a saddle. Sweeping $\hat{M}$ therefore traces a fold with hysteresis (Figure 10a): two stable legitimacy branches joined by an unstable one, colliding at two folds, so history — not payoffs — selects which regime a community occupies. Legitimacy does not change the quantal response itself; it decides where on it the

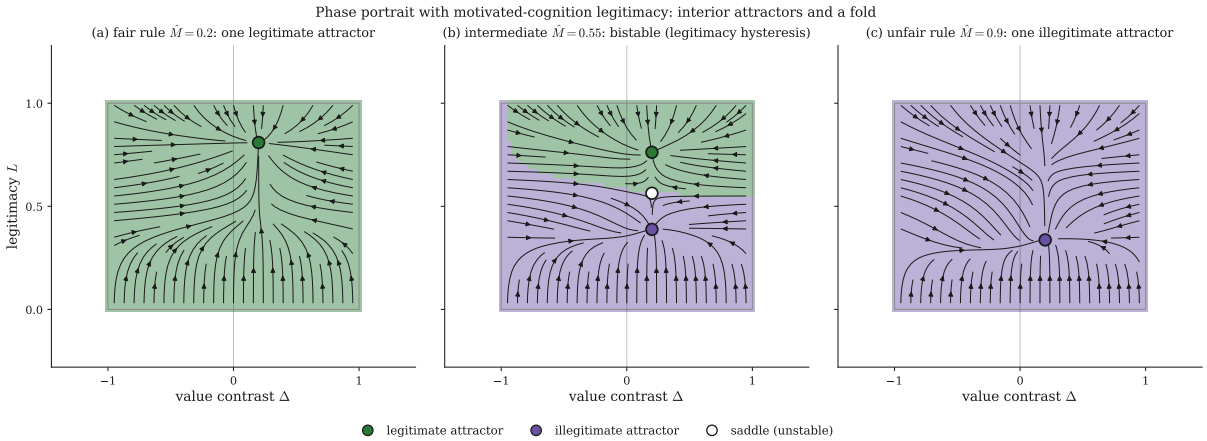



Figure 9: Phase portrait of the mean-field reduction with motivated-cognition legitimacy (7), streamlines over basins of attraction; filled markers are stable attractors, open markers saddles. **(a)** Fair rule ($\hat{M} = 0.2$): a single interior attractor at high legitimacy. **(b)** Intermediate ($\hat{M} = 0.55$): *bistable* — a legitimate and an illegitimate interior attractor separated by a saddle (legitimacy hysteresis). **(c)** Unfair rule ($\hat{M} = 0.9$): a single interior attractor at low legitimacy. The value contrast corrects in every case ($L > 0$); the communities differ in *legitimacy*, not in action.

state settles (Figure 10b). The additive, hard-clipped rule of Section 3 is the singular limit of (7) in which the wells are pushed to the boundaries $L \in \{0, 1\}$; defensive-wrong consensus is then the $L \to 0$ frozen corner.

*Quantal response.* The logistic choice rule is a quantal best response to the augmented value contrast, so a consensus rest point — at which a community's action frequencies reproduce themselves under the choice map — is a quantal-response fixed point in the sense of McKelvey and Palfrey (1995). The regimes are thus stable rest points of a quantal-response dynamic, and legitimacy gating changes which rest point is reached by reweighting the evidence that moves values.

*Fairness preferences.* The legitimacy channel embeds reference-dependent, asymmetric fairness relative to a benchmark, the same primitive as inequity aversion and reciprocity models (Rabin, 1993; Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000); a low-legitimacy rest point is one at which agents' fairness evaluations against the modal benchmark are self-sustaining.

*Set-valued (M) equilibrium.* That the *same* reward rule supports several regimes — legitimate, correct-but-illegitimate, and fairness-polarised (Figure 7) — is naturally read as a set-valued solution. In the M-equilibrium of Goeree and Louis (2021), choices are monotone and beliefs are consequentially unbiased; the fairness-polarisation regime is exactly the case in which community beliefs about the modal agent are *not* consequentially unbiased, so distinct self-consistent (choice, belief) sets coexist.

*Bounded reasoning about a modal agent.* The perceived benchmark $\hat{M}_{i,t}$ is a belief about a representative other to which agents best respond through fairness discomfort — a level-1 / cognitive-hierarchy construct (Nagel, 1995; Stahl and Wilson, 1995; Camerer et al., 2004). Misrepresentation is precisely heterogeneous, incorrect beliefs about that modal agent, and Eq. (6) is a social process by which such beliefs do or do not converge.

*Evolutionary stability.* The corner attractors of the phase portrait are locally stable: a small perturbation of choices or legitimacy decays back (Figure 9). Each is therefore convention-like — an evolutionarily stable rest point of the population dynamic that resists small invasions. The defensive-wrong consensus is the cautionary case: it is a stable convention built on a mis-

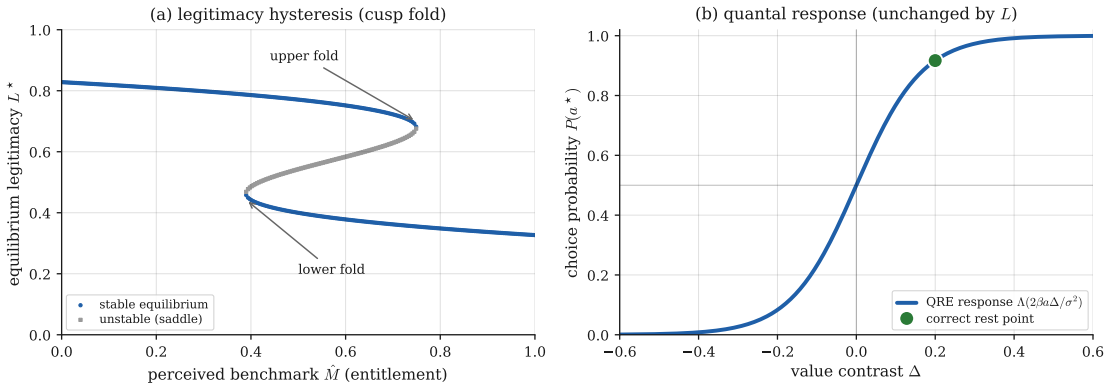



Figure 10: **(a)** Legitimacy hysteresis: the equilibrium legitimacy $L^\star$ versus perceived benchmark $\hat{M}$ is an S-curve — two stable branches (high/low legitimacy) joined by an unstable saddle branch, meeting at two folds (a cusp catastrophe). Between the folds the outcome is history-dependent. **(b)** The quantal-response choice map is unchanged by legitimacy; the gate decides where on it the state settles.

calibrated benchmark, so once a community has entered its basin, restoring accurate feedback does not dislodge it — the feedback is no longer admitted as evidence.

**Proposition 3** (Rest points and their equilibrium reading)**.** *Consider the deterministic community recursion. (i) A symmetric consensus rest point is a quantal-response fixed point of the augmented-value game. (ii) If communities begin with disagreeing benchmarks and permeability and modal transmission are small, the recursion admits distinct community-level rest points (a polarised configuration); reading action frequencies as choices and perceived benchmarks as beliefs, this is an M-equilibrium whose beliefs are not consequentially unbiased across communities.*

(Proof in Appendix A.)

### 6.1 A complete characterisation of the behavioural equilibrium

Write the mean-field map of Section 6 as an operator $T_{\hat{M}}$ on the community state $(Q, L) \in [0,1]^2 \times [0,1]$: choices are the quantal response to the augmented value, and rewards, the legitimacy-gated value update and the expected legitimacy update are taken in expectation. A *behavioural equilibrium* at benchmark $\hat{M}$ is a fixed point $T_{\hat{M}}(Q^\star, L^\star) = (Q^\star, L^\star)$: a self-consistent (choice distribution, values, legitimacy) triple reproduced by gated learning and quantal choice.

**Proposition 4** (Existence and value nullcline)**.** *For every $\hat{M}$, $T_{\hat{M}}$ is continuous on a compact convex set, so a behavioural equilibrium exists (Brouwer). At any equilibrium with $L^\star > 0$ the gated value nullcline forces $Q^\star(a) = \mu_a$ on both arms, so the contrast is $\Delta^\star = \mu_{a^\star} - \mu_o > 0$ and the choice is the quantal response $\Lambda(2\beta a\Delta^\star/\sigma^2)$; only at $L^\star = 0$ can $Q^\star$ be frozen away from the reward means. The value coordinate is therefore monostable whenever legitimacy is positive, and the equilibrium set is indexed by the legitimacy roots of the cusp (7).*

**Proposition 5** (Legitimacy bistability, folds, and hysteresis)**.** *Write $x = L - \frac{1}{2}$ and the tilt $h(\hat{M}) = -\mu_t(\bar{D}^F(\hat{M}) - \tau^F)$, increasing in perceived unfairness. The interior legitimacy equilibria are the roots of $x^3 - (r/s)x - h/s = 0$. For $|h| < h_c := 2s\,(r/(3s))^{3/2}$ there are* three *equilibria — two stable (a legitimate $L^\star_+$ and an illegitimate $L^\star_-$, both interior) separated by an unstable saddle — and for $|h| > h_c$ a single stable equilibrium. As $\hat{M}$ increases through the band where $|h(\hat{M})| < h_c$, the system traverses a bistable region bounded by two saddle-node (fold) bifurcations at $h = \pm h_c$: legitimacy is* hysteretic *(a cusp catastrophe), and within the band the attained*

*regime is selected by history, not by $\hat{M}$ (Figure 10a). A stable equilibrium at $x^\star$ is asymptotically stable because $r - 3s\,x^{\star 2} < 0$ there.*

Under community heterogeneity the picture is additionally set-valued: Proposition 3 and Lemma 1 give that when $\psi\Gamma$ is small the benchmark gap persists and distinct community-level equilibria coexist — an M-equilibrium whose beliefs are not consequentially unbiased across communities (Goeree and Louis, 2021), reducing to a regular quantal response equilibrium (McKelvey and Palfrey, 1995) in the symmetric, single-benchmark case. The additive, hard-clipped legitimacy rule of Section 3 is the singular limit in which the two wells sit at the boundaries $L \in \{0, 1\}$ and the fold degenerates to a boundary transcritical at $\hat{M}^\star = (\tau^F + \nu_R \mu_{a^\star}/[\nu_F(1 - \mu_{a^\star})])/\phi^-$; defensive-wrong consensus is then the frozen $L \to 0$ corner. Proofs of Propositions 4 and 5 are in Appendix A.

**Dynamic-programming formulation.** Each agent's value estimate follows the legitimacy-gated Robbins–Monro recursion $Q_{t+1}(a) = Q_t(a) + \alpha L\,(R - Q_t(a))$ on visits, whose decreasing-step limit is the stochastic-approximation ODE $\dot{Q}(a) = L\,(\mathbb{E}[R \mid a] - Q(a))$ (Borkar, 2008). The associated gated Bellman/TD operator $\mathcal{T}_L Q(a) = Q(a) + L\,(\mathbb{E}[R \mid a] - Q(a))$ is a contraction of modulus $1 - \alpha L$ with unique fixed point $Q^\star(a) = \mathbb{E}[R \mid a]$ when $L > 0$, and the identity map (learning halts) at $L = 0$. Legitimacy is thus a state-dependent weight on the reward-prediction-error backup — a reference-dependent Markov decision process whose effective reward signal is $L\,\delta$, the population-level analogue of a gated prediction-error readout (Joiner et al., 2017). The interaction is a mean-field (population) game: each agent's gated MDP is coupled to the others through the anticipatory field $S$ and the transmitted benchmark $\hat{M}$, and a behavioural equilibrium is the consistency point of quantal best response with gated learning — a mean-field game with a legitimacy-gated Bellman operator, whose equilibrium selection is governed by Proposition 5.

## 7 Discussion

Three claims follow. First, the authority of feedback is a state variable, not a constant: once legitimacy is endogenous, the same reward stream can teach a lot or nothing, and collective error can be defended rather than corrected. Second, fairness benchmarks are social: because the modal-agent benchmark is transmitted on the network, communities can polarise in legitimacy under identical payoffs, and network geometry that spreads information also spreads grievance. Third, epistemic and legitimacy outcomes are independent axes: a society can be right and aggrieved (correct-but-illegitimate) or wrong and self-justifying (defensive wrong). Relative to the credibility baseline (Bontemps and Banerjee, 2026), the new and policy-relevant failure mode is that interventions which restore accurate feedback may not help if that feedback is no longer treated as legitimate.

**Cognitive interpretation of the parameters.** The parameters carry direct psychological readings. The asymmetric discomfort weights $\phi^- > \phi^+$ encode loss-averse, self-serving fairness: shortfalls below the benchmark hurt more than windfalls please, the reference-dependent form of inequity aversion (Fehr and Schmidt, 1999) placed inside a learning loop. The acceptability threshold $\tau^F$ is a tolerance for perceived unfairness before legitimacy erodes; the loss and recovery rates $\nu_F, \nu_R$ are the speed of delegitimation and of its repair; the benchmark transmission rate $\psi$ is susceptibility to socially communicated grievance; and the perceived benchmark $\hat{M}$ is a (possibly distorted) belief about what a representative agent is entitled to. Legitimacy $L$ itself is

the evidential weight the agent grants to reward feedback — a motivated-cognition gain on the prediction error rather than on the choice.

**A normative benchmark.** The natural reference is the *ungated* learner ($L \equiv 1$), for which reward feedback retains full authority — the epistemic ideal in which evidence always teaches. At the collapse operating point the gate is costly in exactly this sense: removing it (ablation A1) lifts the population mass on the optimal action from $m(a^\star) \approx 0.01$ to 0.91 and cuts defensive-wrong consensus from $P = 1.00$ to 0.07 (Table 4). Legitimacy gating is therefore not epistemically neutral: at this point it converts a nearly-correct society into a nearly-certainly-wrong one, a $\sim$90 percentage-point loss of correct-action adoption that measures the welfare cost of illegitimacy. The same gate is protective elsewhere (it lets a community withhold assent from a genuinely unfair rule); the model makes the trade-off explicit rather than assuming feedback is always admitted.

**Testable predictions.** The mechanism yields lab-testable predictions. **(i)** Manipulating perceived fairness *without* changing payoffs — by shifting the communicated modal-agent benchmark — should change how much people learn from identical reward feedback, measurable as a slower value update (smaller learning rate) under low perceived legitimacy. **(ii)** The loss/gain asymmetry $\phi^- > \phi^+$ predicts that below-benchmark outcomes erode legitimacy faster than equal above-benchmark outcomes restore it, an asymmetry recoverable from trial-by-trial legitimacy or acceptance ratings. **(iii)** An acceptability threshold predicts a nonlinear, hysteretic collapse: legitimacy is stable under small unfairness and drops sharply once discrepancy exceeds $\tau^F$. **(iv)** Because benchmarks are transmitted, seeding two weakly connected groups with different fairness benchmarks should produce divergent legitimacy — and even divergent choices — under identical payoffs, with the divergence controlled by inter-group permeability. **(v)** Legitimacy and its benchmark are *latent*: estimate them per trial from acceptance ratings, response times and fairness reports, and test whether the gated-update model (learning rate scaling with inferred legitimacy) predicts behaviour better than an ungated learner. **(vi)** Because the defensive-wrong state is a stable attractor (Figure 9b), restoring accurate feedback to a group already in its basin should *not* correct it, whereas restoring legitimacy — reducing perceived unfairness — should reopen learning: a testable dissociation between "better information" and "fairer rule" interventions. Each maps onto a manipulable quantity in reward-learning or public-goods paradigms with a fairness manipulation (Figure 11).

**Limitations.** The baseline is deliberately narrow: binary actions, stationary rewards, fixed networks, a single fairness benchmark (the modal-agent reward benchmark), and a legitimacy gate on the private prediction error only. It excludes strategic misrepresentation, endogenous rewiring, multiple competing fairness philosophies, institutional rule change, and empirical calibration. These are natural extensions once the minimal mechanism is fixed.

## 8 Conclusion

We have presented a self-contained model of legitimacy-gated social learning in which reward feedback disciplines beliefs only to the extent that the reward rule is treated as legitimate, and legitimacy is itself learned from socially transmitted fairness benchmarks. The model produces legitimate, correct-but- illegitimate, defensive-wrong, and fairness-polarised regimes, and shows that legitimacy collapse turns correctable error into persistent error. A replication package regenerates every result.

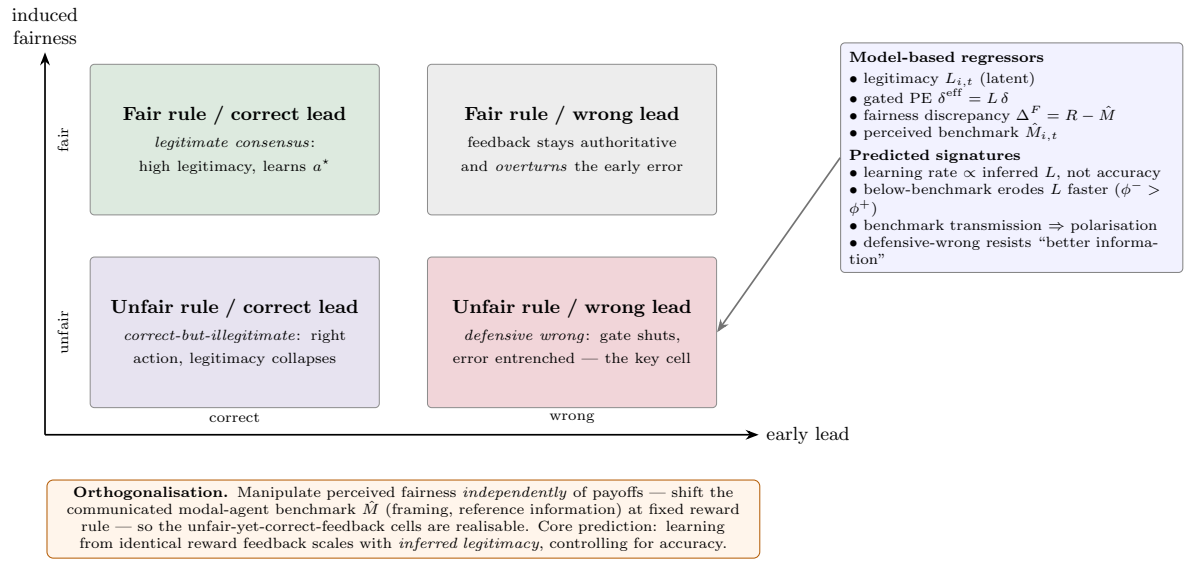



Figure 11: Model-based experimental design. Perceived fairness is manipulated orthogonally to payoffs (by shifting the communicated modal-agent benchmark $\hat{M}$), crossing an early lead (correct vs wrong) to realise the four regimes; the unfair/wrong-lead cell is the falsifiable defensive-wrong prediction. The model supplies latent regressors (legitimacy $L$, gated PE $L\delta$, fairness discrepancy, benchmark $\hat{M}$) and predicted signatures for a model-based analysis.

The contribution sits at the intersection of three mature literatures and adds the connective tissue between them. From *behavioural game theory* it takes the stochastic-choice, boundedly-rational equilibrium tradition — quantal response and its regular form, the anomalies it was built to explain, set-valued M-equilibrium, and level-$k$ reasoning (McKelvey and Palfrey, 1995; Goeree et al., 2005; Goeree and Holt, 2001; Goeree and Louis, 2021; Crawford et al., 2013) — and supplies the missing dynamics: the regimes here are not posited but *selected* by a gated learning process whose rest points we characterise completely (Propositions 4–5) and whose collapse is a transcritical bifurcation. From the *economics of fairness and endogenous norms* it takes reference-dependent, asymmetric fairness and the *heterogeneity* of fairness ideals (Fehr and Schmidt, 1999; Charness and Rabin, 2002; Cappelen et al., 2007; Andre, 2024; Bowles, 1998), and makes the reference a transmissible state, so that grievance itself diffuses on a network. The resulting fairness polarisation is a social analogue of *quasi-non-ergodicity*: like agents who trade on a public signal and "continue to disagree forever" (Bouchaud and Farmer, 2023), communities separated by a benchmark gap settle into path-dependent, self-fulfilling legitimacy regimes that history, not payoffs, selects. From *neuroeconomics and cognitive neuroscience* it takes the view of feedback whose evidential weight is contextually modulated, positioning legitimacy on the state-representation side of the value-versus-state debate over prefrontal computation (Padoa-Schioppa and Conen, 2017; Hayden and Niv, 2021; Joiner et al., 2017) and casting defensive-wrong consensus as socially-induced impairment of reversal learning (Izquierdo et al., 2017). Where the field currently treats the authority of evidence as fixed, the model makes it an endogenous, learnable, transmissible state — and the policy-relevant consequence is sharp: interventions that restore accurate information do not help a community that no longer treats information as legitimate; only interventions that restore legitimacy reopen learning.

# A Proofs

*Proof of Proposition 1.* Take expectations of (3) ignoring the projection (valid in the interior); the two terms are the expected erosion and recovery. The sign of the sum determines the drift. □

*Proof of Proposition 2.* Substituting $\delta_{i,t}^{\text{eff}} = \bar{L}\,\delta_{i,t}$ into (5), the chosen-arm value follows $Q_{t+1} = Q_t + \alpha\bar{L}\,(R - Q_t)$ on visits, a contraction toward $\mathbb{E}[R]$ with modulus $1 - \alpha\bar{L}$; the relaxation time is $-1/\log(1-\alpha\bar{L}) \sim 1/(\alpha\bar{L})$ (Sutton and Barto, 2018; Borkar, 2008). □

*Proof of Lemma 1.* Apply (6) to community means; the antisymmetric mode has multiplier $(1-\psi) + \psi(B_{cc} - B_{cd}) = 1 - 2\psi\Gamma$, using $B_{cd} = 1 - B_{cc} = \Gamma$ for two communities. □

*Proof of Proposition 3.* (i) A rest point is a self-consistent choice distribution under the logistic (quantal) map, hence a quantal-response fixed point (McKelvey and Palfrey, 1995). (ii) The benchmark recursion (6) contracts the antisymmetric mode by the factor $1-2\psi\Gamma$ per trial, which $\to 1$ as $\psi\Gamma \to 0$, so a benchmark gap persists and the two communities settle at different (choice, benchmark) pairs; monotone choices together with community-specific self-consistent beliefs are the defining conditions of M-equilibrium (Goeree and Louis, 2021). Convergence of the finite-$N$ ensemble to these rest points is the standard stochastic-approximation limit (Borkar, 2008). □

*Proof of Proposition 4.* The choice probability is a continuous (logistic) function of the values, the expected gated value increment and the expected legitimacy increment are continuous in $(Q, L)$, and the projection onto $[0,1]$ is continuous; hence $T_{\hat{M}}$ maps the compact convex set $[0,1]^2 \times [0,1]$ continuously into itself and Brouwer's theorem gives a fixed point. At a fixed point the value equation reads $L^\star\,\mathbf{1}\{A = a\}\,(\mathbb{E}[R \mid a] - Q^\star(a)) = 0$ for each arm; if $L^\star > 0$ this forces $Q^\star(a) = \mu_a$ on visited arms, so $\Delta^\star = \mu_{a^\star} - \mu_o$ and the choice is the quantal response to $\Delta^\star$, while at $L^\star = 0$ the increment vanishes and $Q^\star$ may be frozen. The value coordinate is thus monostable for $L^\star > 0$, and the equilibria are indexed by the legitimacy roots analysed in Proposition 5. □

*Proof of Proposition 5.* At the value nullcline the choice distribution is fixed, so $\bar{D}^F(\hat{M})$ and hence the tilt $h(\hat{M})$ are constants and (7) is an autonomous scalar cubic. Interior equilibria solve $g(x) := x^3 - (r/s)x - (h/s) = 0$. The depressed cubic $x^3 + px + q$ with $p = -r/s < 0$, $q = -h/s$ has three distinct real roots iff its discriminant is positive, $4p^3 + 27q^2 < 0$, i.e. $(h/s)^2 < \frac{4}{27}(r/s)^3$, equivalently $|h| < h_c = 2s\,(r/(3s))^{3/2}$; it has one real root when $|h| > h_c$ and a double root at $|h| = h_c$. A scalar rest point $x^\star$ is asymptotically stable iff $\dot{x}'(x^\star) = r - 3s\,x^{\star 2} < 0$; the two outer roots satisfy this and the middle root does not, giving two stable wells and one saddle in the bistable band. As $\hat{M}$ varies, $h(\hat{M})$ crosses $\pm h_c$, where a stable root and the saddle coalesce and annihilate — a saddle-node (fold). The two folds bound a bistable interval over which the attained equilibrium depends on the initial condition, which is hysteresis (the cusp normal form). Finally, letting the wells' curvature diverge so that the stable roots are driven to the boundaries recovers the additive, hard-clipped limit, in which the pair of folds degenerates to the single boundary transcritical at $\hat{M}^\star = (\tau^F + \nu_R\mu_{a^\star}/[\nu_F(1-\mu_{a^\star})])/\phi^-$ (the balance point of Proposition 1). □

# B ODD+D description

**A.1 Purpose and patterns.** To determine when reward feedback stops being treated as valid evidence because the rule is perceived as unfair, and how network geometry shapes the

result. Target patterns: **P1** legitimate consensus; **P2** correct-but-illegitimate consensus; **P3** defensive wrong consensus (legitimacy collapse making a correctable error persistent); **P4** fairness polarisation under identical payoffs; **P5** an acceptability-threshold transition; **P6** geometry-controlled benchmark diffusion. Not targeted: empirical fairness time-series, continuous actions, non-stationary rules.

**A.2 Entities, state variables, scales.** Agents (values $Q_{i,t}(a)$, confidence $C_{i,t}$, legitimacy $L_{i,t}$, perceived benchmark $\hat{M}_{i,t}$), communities, two arms, and a row-stochastic coupling matrix $B$. Time is discrete at the trial level with horizon $T$.

**A.3 Process overview and scheduling.** Synchronous trials with a simultaneous update: (1) credibility field from the previous period; (2) augmented values; (3) process-based choice and confidence; (4) reward; (5) fairness discrepancy, discomfort and legitimacy update from the pre-update state; (6) legitimacy-gated private plus retrospective-social value update; (7) modal-agent benchmark update; (8) record.

**A.4 Design concepts.** *Basic principles:* prediction-error learning gated by legitimacy; fairness judged against a social benchmark. *Emergence:* regimes from rules and coupling. *Adaptation:* gated value revision and legitimacy/benchmark learning. *Objectives:* myopic reward through learned values. *Sensing:* neighbours' actions, confidence and benchmarks; own reward vs benchmark. *Interaction:* weighted by $B$ and credibility. *Stochasticity:* choice and Bernoulli rewards. *Collectives:* communities. *Observation:* the observables of Section 4.

**A.5 Initialisation.** Community-constant initial values, legitimacy $L_0$ and benchmarks $\hat{M}_0$ per Table 5 and the per-experiment scenarios; benchmark disagreement is set through $\hat{M}_0 = (0.5 \mp |\Delta\hat{M}_0|/2)$.

**A.6 Input data.** None; the model is synthetic.

**A.7 Submodels.** Credibility field; augmented value and choice; confidence; fair benchmark $r^F = f(a, \hat{M})$; fairness discrepancy (1); asymmetric discomfort (2); legitimacy update (3); legitimacy-gated prediction error (4); value update (5); modal-agent transmission (6); regime classifier. Each maps to code and a test in the ODD-to-code map (`ODD_to_code_map.md`).

**A.8 Parameters.** Global defaults in Table 5; per-experiment values in `scenarios.py`.

**A.9 Code-location map.** See `ODD_to_code_map.md`: each equation maps to a function in `code/legitimacy/` and a check in `tests/test_legitimacy.py`.

**A.10 Verification and reproducibility.** Appendix C and the replication metadata; all results regenerate from seeded scripts.

# C Verification

The implementation passes an automated suite of 11 checks (all green): legitimacy and value bounds in $[0, 1]$; the effective-prediction-error identity $\delta^{\text{eff}} = L\delta$ (and $L \equiv 1$ when the gate is off); the discomfort asymmetry ($\phi^- > \phi^+ \Rightarrow$ larger legitimacy loss for disadvantageous unfairness); the acceptability-threshold partition; modal-agent misperception (a benchmark gap

Table 5: Global default parameters (`code/legitimacy/config.py`); per-experiment values are in `scenarios.py`. Values are projected to $[0, 1]$; the fairness-in-choice term is off by default.

| Symbol | Default | Role | Config key |
|---|---|---|---|
| $\mu_1, \mu_2$ | 0.6,0.4 | arm reward means | `mu` |
| $\lambda$ | 0.5 | anticipatory weight | `lam` |
| $\eta$ | 0.2 | retrospective weight | `eta` |
| $\beta, \sigma, a$ | 6.0,1.0,1.0 | DDM choice | `beta,sigma,a_thr` |
| $\alpha$ | 0.2 | private learning rate | `alpha` |
| $\nu_F, \nu_R$ | 0.05,0.01 | legitimacy loss / recovery | `nu_F,nu_R` |
| $\tau^F$ | 0.3 | acceptability threshold | `tau_F` |
| $\phi^-, \phi^+$ | 1.0,0.25 | discomfort weights ($\phi^- > \phi^+$) | `phi_minus,phi_plus` |
| $\psi$ | 0.1 | modal-agent transmission | `psi` |
| $L_0$ | 1.0 | initial legitimacy | `L0` |

persists under low permeability and erodes under full mixing); modal-agent bounds (including idiosyncratic noise); exactness of the balanced-block community field for the modal-agent update; the regime classifier on hand-built inputs; seed reproducibility; the no-gate ablation (removing the gate restores correction of an early wrong lead); and the true-benchmark ablation (pinning $\hat{M} = M$, and the effort-reward benchmark form).

# D Parameters

Per-experiment scenario parameters (population size, horizon, replications, $\lambda$, $\eta$, $\nu_F$, $\nu_R$, $\tau^F$, $\hat{M}_0$, initial values and seeds) are defined in `code/legitimacy/scenarios.py`, the authoritative source, and summarised in `reports/replication_audit.md`.

# E Robustness

The qualitative regimes are stable to the terminal threshold and to moderate changes in the discomfort weights and loss/recovery rates around the values in Table 5; the legitimacy-collapse boundary follows Proposition 1. A systematic robustness sweep (threshold, horizon, benchmark strength, permeability, and a graph-perturbation check analogous to the credibility baseline) is the immediate next step and is scaffolded in `reports/robustness_summary.md`.

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